

Exercise 7.1: Epistemic vs. Aleatoric Uncertainty

Define epistemic and aleatoric uncertainty.

1. Which type is more relevant to OOD inputs (e.g., night images for a model trained on day images)?

2. Which type would you expect to be dominant in a correctly classified, in-distribution image?

- Epistemic uncertainty is uncertainty caused by a lack of knowledge or insufficient training data. It can be reduced by collecting more data or improving the model.
- Aleatoric uncertainty is uncertainty caused by inherent noise or ambiguity in the data. It cannot be eliminated even with more data.

Epistemic uncertainty is more relevant for OOD inputs because the model has not seen similar examples during training and therefore lacks knowledge about them.

Aleatoric uncertainty would be dominant because the model is already familiar with the data distribution, leaving mainly uncertainty caused by noise or ambiguity in the input.

Exercise 7.2: Calibration and ECE Explain what it means for a classifier to be well-calibrated. Define the Expected Calibration Error (ECE) and describe how it is computed.

Calibration measures whether a model's confidence can be trusted

- A classifier is well-calibrated when its predicted confidence matches the true probability of being correct. For example, predictions made with 80% confidence should be correct approximately 80% of the time.

How far confidence is from reality?

- Expected Calibration Error (ECE) measures the difference between predicted confidence and actual accuracy. Predictions are grouped into confidence bins, and for each bin the absolute difference between average confidence and accuracy is calculated. ECE is the weighted average of these differences across all bins.

Exercise 7.3: Cost-Optimal Downstream Decisions

A pedestrian detector outputs $p = p(\text{ped} \mid x) \in [0, 1]$. The autopilot must choose between two actions with asymmetric costs:

	pedestrian present	no pedestrian
BRAKE	0	$C_{\text{FP}} = 1$
CONTINUE	$C_{\text{FN}} = 100$	0

At which p should the autopilot brake in order to minimize the costs?

1. Write down the expected loss (costs) of each action as a function of p : $\mathbb{E}[\mathcal{L} \mid \text{BRAKE}]$ and $\mathbb{E}[\mathcal{L} \mid \text{CONTINUE}]$.
2. Find the threshold τ^* at which both actions have equal expected loss. For $p > \tau^*$ the autopilot should brake; for $p < \tau^*$ it should continue.
3. Compute τ^* for the costs above. How does it compare to the standard argmax rule ($\tau = 0.5$)?
4. Why does τ^* only yield the optimal decision when p is well-calibrated? What can go wrong when the same threshold is applied to an overconfident model?

1. Expected Loss:

$$\begin{aligned} \mathbb{E}[\mathcal{L} \mid \text{brake}] &= 1 - p \\ \mathbb{E}[\mathcal{L} \mid \text{continue}] &= 100p \end{aligned}$$

2. Decision Threshold :

$$\begin{aligned} 1 - p &= 100p \\ p &= 1/101 \end{aligned}$$

3. Compute τ^* :

$$\tau^* = 1/101 = 0.0099$$

This is much lower than the standard threshold of 0.5 because the cost of missing a pedestrian is much higher than the cost of unnecessary braking

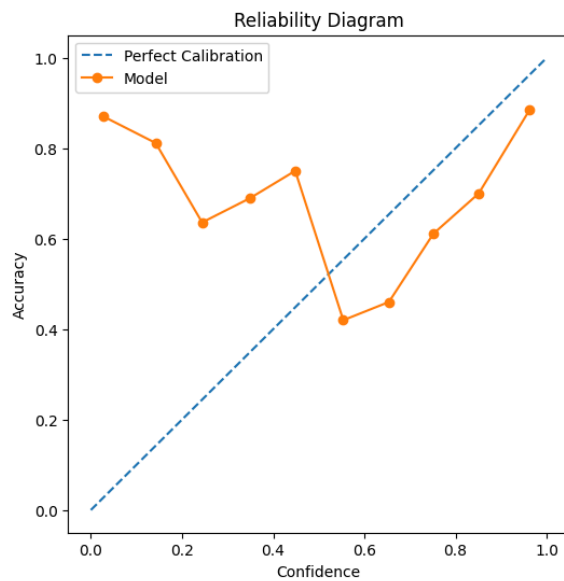
4. The threshold τ^* assumes that the predicted probability accurately reflects the true likelihood of a pedestrian. If the model is overconfident or underconfident, the calculated threshold may lead to unsafe decisions because the probabilities no longer correspond to reality.

Exercise 7.4: Measuring Calibration

1. Compute the ECE of each of your three trained CARLA models on the in-distribution test set.

2. Plot a reliability diagram (confidence vs. accuracy) for each model.

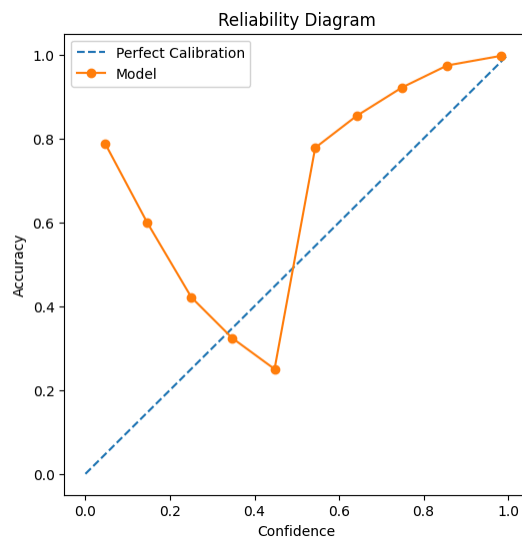
Pedestrian :



ECE = 0.0992

- Moderate calibration error.
- Confidence does not perfectly match accuracy.
- Slightly overconfident in several confidence ranges.

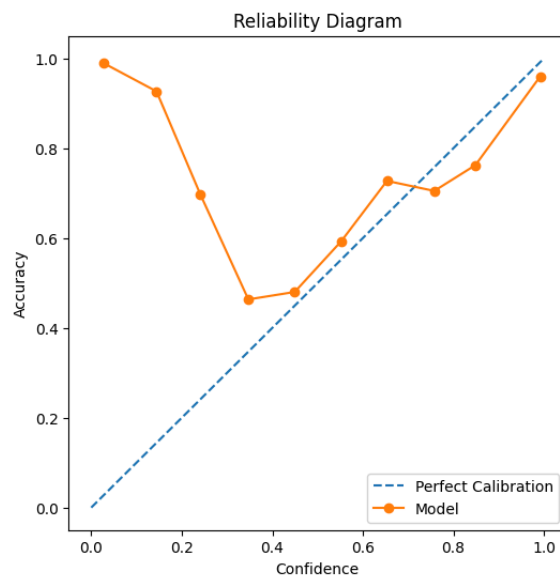
Vehicle:



ECE = 0.1069

- Highest ECE among the three models.
- Most poorly calibrated model.
- Shows both overconfidence and underconfidence depending on the confidence region.

Traffic:



ECE = 0.0350

- Best calibrated model.
- The reliability curve stays close to the diagonal.
- Confidence scores are generally trustworthy.

3. Are your models over- or underconfident? Does this pattern hold consistently across all three models?

The Pedestrian model is mostly overconfident, the Vehicle model shows mixed calibration behaviour, and the Traffic Light model is the best calibrated. Therefore, there is no single calibration pattern that holds consistently across all three models.

Exercise 7.5: Temperature Scaling Temperature scaling adjusts the model's logits as:

$$p(y | x) = \text{softmax } f(x) / T$$

1. Apply temperature scaling to each of your three models. Optimize T on the validation set using negative log-likelihood.
2. Report the ECE before and after scaling for each model.

Model	Best T	ECE Before	ECE After
Pedestrian	1.95	0.0992	0.0362
Vehicle	1.47	0.1069	0.1056
Traffic Light	1.52	0.0350	0.0346

- Temperature scaling substantially improved the calibration of the Pedestrian model.
- The Vehicle model showed almost no improvement after scaling.

- The Traffic Light model was already well calibrated, so only a small improvement was observed.
- All optimal temperature values were greater than 1, indicating that the models tended to be overconfident.

Exercise 7.6: Cost-Optimal Decision in Practice Use the costs from Exercise 3 (CFN = 100, CFP = 1, $\tau^* \approx 0.0099$, i.e. a predicted probability of roughly 1 %) and apply them to your pedestrian classifier.

1. On the in-distribution test set, compute the total loss $L = \text{CFN} \cdot \text{\#FN} + \text{CFP} \cdot \text{\#FP}$ for your uncalibrated model at both $\tau = 0.5$ and $\tau = \tau^*$.

Model	Threshold	Loss	FP	FN
Uncalibrated	$\tau = 0.5$	54607	107	545
Uncalibrated	$\tau = 0.0099$	9515	2215	73
Calibrated	$\tau = 0.5$	54607	107	545
Calibrated	$\tau = 0.0099$	2904	2904	0

2. Repeat for your temperature-scaled model at both thresholds.

3. Arrange the four results in a 2×2 table (rows: uncalibrated / calibrated; columns: $\tau = 0.5$ / $\tau = \tau^*$). Which combination gives the lowest total loss?

	$\tau = 0.5$	$\tau = 0.0099$
Uncalibrated	54607	9515
Calibrated	54607	2904

- The calibrated model with $\tau = 0.0099$ gives the lowest total loss: Loss = 2904
- The calibrated model with the cost-optimal threshold $\tau = 0.0099$ achieved the lowest total loss (2904). Lowering the threshold significantly reduced false negatives, which are much more expensive than false positives (CFN = 100 vs CFP = 1). Temperature scaling further improved the decision-making process by producing better-calibrated probabilities, allowing the cost-optimal threshold to be more effective.

Exercise 7.7: Tracing Overconfidence Through the Safety Analysis You have now measured how miscalibrated confidence propagates silently to downstream decisions. In STPA, an overconfident misclassification is a causal factor: a mechanism that explains why an existing unsafe control action occurs. The goal of this exercise is to add it as a loss scenario and trace it through to a verifiable constraint.

1. Causal scenario. Revisit your causal loss scenarios from Exercise 2.8. The hazard (“vehicle fails to brake for a pedestrian”, Exercise 2.4) and the unsafe control action (“the planner does not command braking while a pedestrian is in the path”, Exercise 2.6) already exist - you do not need new ones. Add a causal scenario describing a new way that UCA arises: the pedestrian classifier produces a false negative yet reports high confidence, so the planner registers “no pedestrian, reliable” and no low-confidence fallback is triggered.

Hazard:

The vehicle fails to brake for a pedestrian.

UCA:

The planner does not command braking while a pedestrian is in the vehicle's path.

New Causal Scenario:

The pedestrian classifier produces a false negative prediction while reporting high confidence. The planner interprets this prediction as "no pedestrian detected and prediction reliable" and therefore does not activate any fallback behaviour. As a result, the vehicle continues moving and fails to brake for the pedestrian.

2. Safety constraint. From the scenario, derive at least one new safety constraint of each kind (cf. Exercise 2.7):

- **a model-level constraint on calibration, expressed with the ECE you measured; and**
- **a system-level constraint on how the planner uses the confidence score - e.g. the fallback when confidence is low, or adopting the cost-optimal threshold τ^* from Exercise 3 instead of 0.5.**

a) Model-Level Constraint (Calibration):

- The pedestrian classifier shall maintain an Expected Calibration Error (ECE) below 0.05 on the validation dataset to ensure that confidence scores accurately reflect prediction reliability.

Justification:

- Before temperature scaling:
 - ECE = 0.0992
- After temperature scaling:
 - ECE = 0.0362

The calibrated model satisfies this constraint.

b) System-Level Constraint

- The planner shall not rely solely on a 0.5 classification threshold. It shall use the cost-optimal threshold $\tau^* = 0.0099$ and trigger a safe fallback action whenever confidence is low or uncertainty is detected.

Justification:

The calibrated model with $\tau^*=0.0099$ achieved the lowest loss.

3. Verification. What evidence verifies the model-level constraint? You produced exactly this evidence on this sheet (the ECE before and after temperature scaling, Exercise 7.4–Exercise 7.5). Does your calibrated model meet the threshold you set?

The model-level constraint is verified using the ECE measurements from Exercises 7.4 and 7.5.

Model State	ECE
Before Temperature Scaling	0.0992
After Temperature Scaling	0.0362

Yes.

The calibrated pedestrian model achieved:

$ECE = 0.0362$

which is below the proposed threshold:

$ECE < 0.05$

Therefore, the calibration constraint is satisfied.

4. Residual risk. Even with a perfectly calibrated model, what does the constraint not cover? Does calibration alone close the scenario, or is the system-level fallback still required?

Even with perfect calibration, the classifier can still make incorrect predictions.

Calibration only ensures that confidence scores are trustworthy; it does not guarantee correctness.

For example:

Pedestrian Present

↓

Classifier predicts No Pedestrian

↓

Confidence = 60%

The prediction can still be wrong even if the confidence is perfectly calibrated.

Therefore:

- Calibration alone does not eliminate the hazard.
- The system-level fallback remains necessary.
- Safe responses such as braking, slowing down, or using the cost-optimal threshold must still be implemented to mitigate remaining risk.